# TEXT GENERATION USING LSTM NETWORKS

```
import tensorflow as tf
import numpy as np

# Step 1: Download and Prepare Dataset
text = tf.keras.utils.get_file(
    'shakespeare.txt',
    'https://raw.githubusercontent.com/karpathy/char-
rnn/master/data/tinyshakespeare/input.txt'
)
text = open(text, 'r').read().lower()
chars = sorted(set(text))
c2i = {c: i for i, c in enumerate(chars)}
i2c = {i: c for i, c in enumerate(chars)}

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# Step 2: Create Input and Output Sequences
seq_len = 40
X = []
y = []
for i in range(len(text) - seq_len):
    input_seq = text[i:i + seq_len]
    target_char = text[i + seq_len]
    X.append([c2i[c] for c in input_seq])
    y.append(c2i[target_char])

X = np.array(X)
y = np.array(y)

# Step 3: Build the LSTM Model
model = tf.keras.Sequential([
    tf.keras.layers.Embedding(len(chars), 64, input_length=seq_len),
    tf.keras.layers.LSTM(128),
    tf.keras.layers.Dense(len(chars), activation='softmax')
])

model.compile(loss='sparse_categorical_crossentropy', optimizer='adam')
model.fit(X, y, batch_size=128, epochs=1)

# Step 4: Define the Text Generation Function
```

```
def generate(seed, length=300):
    seq = [c2i[c] for c in seed.lower()]
    for _ in range(length):
        inp = np.array(seq[-seq_len:]).reshape(1, -1)
        pred = model.predict(inp, verbose=0)[0]
        next_idx = np.random.choice(len(pred), p=pred)
        seq.append(next_idx)
    return seed + ''.join(i2c[i] for i in seq[len(seed):])

# Step 5: Generate and Display Text
print("\nGenerated Text:\n")
print(generate("shall i compare thee to a summer's day?\n"))
```

OUTPUT:

```
Downloading data from https://raw.githubusercontent.com/karpathy/char-rnn/master/data/tinyshakespeare/input.txt
1115394/1115394 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step

/usr/local/lib/python3.13/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argument `input_length` is deprecated. Just remove it.
  warnings.warn(

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Generated Text:

shall i compare thee to a summer's day?

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```

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